

INTRODUCCION A LA CRIPTOGRAFIA MODERNA

Homework 2

Depto. de Computación, FCEN, UBA, 1er. Bimestre 2026

14/4/26

Student:

Due: 23/4/25, 23:59 hs

Instructions

- Upload your solution to Campus; make sure it's only one file, and clearly write your name on the first page. Name the file '`<your last name>_HW2.pdf`.'
- If you are proficient with $\text{\LaTeX}$, you may also typeset your submission and submit in PDF format. To do so, uncomment the `"%\begin{solution}"` and `"%\end{solution}"` lines and write your solution between those two command lines.
- Your solutions will be graded on *correctness* and *clarity*. You should only submit work that you believe to be correct.
- You may collaborate with others on this problem set. However, you must **write up your own solutions** and **list your collaborators and any external sources (including ChatGPT and similar generative AI chatbots)** for each problem. Be ready to explain your solutions orally to a member of the course's staff if asked.

This assignment contains 6 questions, for a total of 60 points.

1. Let F be a pseudorandom permutation. For each of the following encryption schemes (where the key is a uniform $k \in \{0,1\}^n$), show how to decrypt and then state whether the scheme is CPA-secure. Explain your answers.
 - (a) (5 points) To encrypt $m \in \{0,1\}^n$, choose uniform $r \in \{0,1\}^n$ and output the ciphertext $\langle r, F_k(m) \oplus r \rangle$.
 - (b) (5 points) To encrypt $m \in \{0,1\}^n$, choose uniform $r \in \{0,1\}^n$ and output the ciphertext $\langle r, F_r(k) \oplus m \rangle$.
2. (10 points) Show that the encryption scheme presented in Lecture 5 (p. 21) is CCA-secure.
3. Prove that the following modifications of basic CBC-MAC **do not** yield a secure MAC (even for fixed-length messages):
 - (a) (5 points) Mac outputs all blocks $t_1, \dots, t_\ell$, rather than just t_ℓ . (Verification only checks whether t_ℓ is correct.)
 - (b) (5 points) A random initial block is used each time a message is authenticated. That is, choose uniform $t_0 \in \{0,1\}^n$, run basic CBC-MAC over the "message" $t_0, m_1, \dots, m_\ell$, and output the tag $\langle t_0, t_\ell \rangle$. Verification is done in the natural way.

4. (a) (5 points) Provide formal security definitions for *second preimage resistance* and *preimage resistance* of hash functions, similar to the definition of collision resistance we saw in Lecture 6.
(b) (5 points) Prove that any hash function that is collision resistant is second-preimage resistant, and any hash function that is second-preimage resistant is preimage resistant.
5. (10 points) Show how to find a collision in the Merkle tree construction if t is not fixed. Specifically, show how to find two sets of inputs $x_1, \dots, x_t$ and $x'_1, \dots, x'_{2t}$ such that $\mathcal{MT}_t(x_1, \dots, x_t) = \mathcal{MT}_{2t}(x'_1, \dots, x'_{2t})$.
6. (10 points) Before HMAC, it was common to define a MAC for arbitrary-length messages by $\text{Mac}_{s,k}(m) = H^s(k\|m)$, where H is a collision resistant hash function. Show that this is never a secure MAC when H is constructed via the Merkle-Damgård transform. (Assume the hash key s is known to the attacker, and only k is kept secret.)